

I taught Computer Science at the [University of Central Oklahoma](#) for 10 years. During this time I taught an introductory course in PC Assembly Language programming. I grew frustrated at teaching 16-bit real mode programming and decided to change to 32-bit protected mode. However, I soon ran into a problem. I could not find a textbook that covered 32-bit protected mode assembly programming! So, I decided to write my own.

I also did not want students to have to go out and buy expensive software for the course. I decided to base the course on the free *NASM* (Netwide Assembler) and the free GNU *gcc* compiler (however, any 32-bit C compiler would work). Another advantage of these choices was that students could use *Windows*, *Linux* or *FreeBSD* to develop on. (In fact, I use Linux for my main development platform.)

Over one summer I wrote the bulk of this textbook and developed the examples using *LaTeX*. I made a feeble attempt to get the book published and then decided to publish it myself online for free. Why? To return something back to the developer community. I have used many *open source* products and wanted to make a modest contribution of my own.

The book has extensive coverage of interfacing assembly and C code and so might be of interest to C programmers who want to learn about how C works under the hood. All the examples use the free NASM (Netwide) assembler. The tutorial only covers programming under 32-bit protected mode and requires a 32-bit protected mode compiler. It is possible to use the book and examples on a 64-bit OS; however, the programs will be running in 32-bit mode and cannot use any 64-bit functionality.

I have example code files for: DJGPP, Borland, Microsoft, Open Watcom and Linux C compilers. The examples in the text of the tutorial are for DJGPP only, but how to interface with the other compilers is discussed as well. The example files also include macros that allow easy input/output and debugging (register dumps, memory dumps, coprocessor dumps and stack dumps). If you plan on running the examples in the tutorial, you *must* download the appropriate example code file below. It contains support files used by the examples in the tutorial (such as *asm_io.inc*).

Table of Contents

1. Introduction
2. Basic Assembly Language
3. Bit Operations
4. Subprograms
5. Arrays
6. Floating Point
7. Structures and C++

Feedback is welcome. Below are links to the files:

Example code:

- [Linux examples](#)
- [DJGPP examples](#)
- [cygwin examples](#)
- [MS C examples](#)
- [Borland examples](#)
- [Open Watcom examples](#)

Book

- [English PDF File](#)
- [Physical book in English from Lulu](#)
- [French PDF file](#) (Translated by Sébastien Le Ray)
- [German PDF file](#) (Translated by Ulrich Bicheler)
- [Spanish PDF file](#) (Translated by Leonardo Rodríguez Mújica)
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This is the github page of Paul Carter. I'm a software developer using mostly C++, Java and Python.